

SCOTT AFB, IL, USA

WMO: 724338

Lat: 38.550N

Lon: 89.850W

Elev: 459

StdP: 14.45

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13802

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.6	11.7	-2.2	5.0	9.1	2.8	6.5	13.9	26.6	34.8	24.0	33.3	9.1	330	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	95.0	77.7	92.6	76.9	90.2	76.1	80.8	89.7	79.3	88.3	78.0	86.6	8.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.6	151.1	85.3	77.0	142.8	84.1	75.3	135.1	82.6	44.7	89.7	43.1	87.9	41.7	86.6	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.7	19.6	17.4	-1.6	99.8	5.7	5.4	-5.7	103.6	-9.0	106.8	-12.3	109.8	-16.4	113.7		
(5)			-1.5	83.3	6.1	1.7	-5.9	84.5	-9.5	85.5	-13.0	86.4	-17.4	87.6	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.3	30.9	36.7	45.6	56.4	66.4	75.0	78.2	76.9	69.5	58.3	45.3	35.5		
	DBStd	18.60	11.71	10.96	10.85	9.11	7.84	5.93	5.32	5.13	7.09	8.82	9.44	10.12		
	HDD50	1926	600	390	214	40	1	0	0	0	0	24	198	459		
	HDD65	4609	1056	791	605	282	82	6	0	1	35	246	591	914		
	CDD50	4237	9	19	78	232	509	750	873	835	584	280	58	10		
	CDD65	1445	0	0	4	24	125	306	408	371	169	37	1	0		
	CDH74	14272	0	3	48	252	1078	2990	4312	3575	1625	372	17	0		
	CDH80	5862	0	0	4	49	319	1251	1970	1552	622	94	1	0		
	WSAvg	7.3	8.9	9.0	9.1	9.1	7.2	6.3	5.3	5.0	5.2	6.7	7.9	8.2		
	PrecAvg	43.8	2.6	2.3	3.3	4.8	5.2	4.8	3.9	3.7	3.4	3.2	3.7	2.9		
(15) Precipitation	PrecMax	56.7	8.2	4.8	10.0	11.3	15.0	9.6	7.3	6.4	9.3	11.4	7.1	8.1		
	PrecMin	31.2	0.4	0.6	1.0	1.8	0.7	1.2	0.6	1.3	0.4	0.6	0.8	0.7		
	PrecStd	6.5	1.7	1.1	1.7	2.3	3.2	2.4	1.8	1.4	2.2	2.1	1.9	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.9	72.0	79.4	85.6	90.2	95.9	99.6	97.8	94.3	87.5	76.6	66.2		
		MCWB	59.5	59.7	64.0	67.6	73.1	76.6	77.6	77.9	73.6	71.2	62.8	61.0		
	2%	DB	59.6	64.7	73.4	80.9	86.7	92.8	95.3	94.5	90.7	82.2	70.0	61.1		
		MCWB	54.0	56.9	60.3	65.5	71.9	76.2	78.8	77.9	74.2	66.8	59.4	57.0		
	5%	DB	54.2	59.9	68.5	76.6	83.8	90.4	92.7	91.1	87.4	78.3	65.2	56.3		
		MCWB	50.0	53.1	57.9	63.6	70.6	75.6	78.1	77.1	72.6	65.3	58.1	51.7		
	10%	DB	47.9	54.4	63.6	72.5	80.7	87.7	89.9	88.3	83.4	73.4	61.2	50.6		
		MCWB	43.5	48.3	55.3	61.5	69.0	74.3	77.0	75.6	70.0	63.1	54.9	46.5		
	0.4%	WB	61.1	63.0	66.8	71.3	77.3	80.9	83.1	82.7	78.4	74.0	66.4	62.9		
		MCDB	64.3	67.1	75.2	80.5	84.9	89.5	91.4	90.9	88.1	83.3	71.0	65.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	55.6	59.0	63.4	68.5	74.5	78.8	81.1	80.6	76.7	70.9	62.9	58.1		
		MCDB	58.1	63.9	70.0	76.2	82.7	88.4	90.1	89.6	85.7	78.1	66.4	60.4		
	5%	WB	49.7	53.9	60.3	66.0	72.7	77.3	79.7	78.8	74.9	68.1	59.8	52.5		
		MCDB	53.0	58.3	66.3	73.0	80.3	86.5	88.6	87.6	82.6	74.1	64.7	55.0		
	10%	WB	42.9	47.9	56.7	63.6	70.9	75.9	78.3	77.4	72.9	65.4	55.5	46.8		
		MCDB	47.1	54.0	62.8	69.9	77.9	84.5	86.9	85.6	79.9	72.3	60.2	49.6		
	5% DB	MDBR	16.7	18.1	20.4	21.5	20.0	19.9	19.6	20.4	22.8	22.7	19.3	16.2		
		MCDBR	24.6	27.0	27.3	26.2	23.1	23.0	22.2	23.3	26.1	27.2	25.7	23.2		
		MCWBR	19.0	19.2	16.0	14.3	11.3	9.1	8.6	9.5	10.7	13.7	16.9	18.1		
		MCWBR	22.3	24.7	23.3	21.8	19.9	20.0	20.1	20.7	21.2	22.0	21.2	20.1		
(40) Clear-Sky Solar Irradiance	taub		0.308	0.318	0.352	0.403	0.437	0.444	0.472	0.465	0.410	0.353	0.313	0.302		
		taud	2.466	2.424	2.394	2.248	2.212	2.223	2.147	2.160	2.296	2.460	2.513	2.497		
	Ebn at Noon		277	288	288	278	269	266	258	256	265	272	273	271		
		Edh at Noon	26	30	35	43	45	45	48	46	38	29	24	23		
(44) All-Sky Solar Radiation	RadAvg		646	877	1187	1570	1760	2023	1974	1779	1501	1087	735	542		
	RadStd		70	93	115	91	156	140	135	101	98	116	78	56		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140

Nomenclature: See separate page